AWS & Testing Services Budget Estimate

Region: Europe (Stockholm) eu-north-1 | Currency: USD ($)

1. Worst-Case Infrastructure & Hidden Cost Table (24/7 Run)

Service / Surcharge	Specs / Cause	Monthly Cost
AWS NAT Gateway	Private subnet outbound gateway	$32.85
NAT Data Processing	Outbound traffic (PDF & API calls)	$4.50
Public IPv4 Charge	Public IP surcharge ($0.005/hr)	$3.65
EC2 EBS Storage	30 GB GP3 EBS volume	$2.40
RDS Database Compute	db.t3.micro MySQL instance	$12.41
RDS DB Storage	20 GB GP3 storage volume	$2.30
RDS Backups	Automated database backups	$1.00
AWS Secrets Manager	Credentials & API key storage	$0.80
CloudWatch Logs	Log ingestion & storage	$1.50
Cross-AZ Data Transfer	EC2-to-RDS Cross-AZ traffic	$2.00
EC2 Compute	t3.micro Linux server instance	$7.59
Route 53 Hosted Zone	DNS hosted zone (codealgo.live)	$0.50
WORST-CASE INFRA TOTAL	Fully configured production network	**~$71.50 / mo**

2. Page-Based Testing Cost Scaling Table

Page Volume	Vectors Count	DB Space	Gemini Cost	Total (Payg)
100 Pages	~300	~1 MB	$0.02	**$0.02**
1,000 Pages	~3,000	~10 MB	$0.20	**$0.20**
10,000 Pages	~30,000	~100 MB	$2.03	**$2.03**
40,000 Pages	~120,000	~400 MB	$8.10	**$8.10**

3. Cost Optimization Recommendations

1. Remove NAT Gateway: Launch EC2 in a public subnet to save ~$33/mo immediately.
2. Match Availability Zones: Keep EC2 and RDS in the same AZ (e.g. eu-north-1a) to avoid cross-AZ transfer fees.
3. Schedule Compute Hours: Shut down EC2/RDS during nights & weekends to save ~60% of compute fees.
4. Leverage Free Tiers: Keep testing within the 100k vector Pinecone limit and Vercel hobby plan ($0.00).